

Pavithra Venkat Sridhar

Evanston, IL | pavithravenkatsridhar2027@u.northwestern.edu | linkedin.com/in/pavi-venk | pavi-venk.github.io/portfolio

EDUCATION

Northwestern University, USA | Bachelor of Science in Mechanical Engineering **Sept. 2023 - Present**

Robotics concentration, Segal Design Certificate

Coursework: Mechanical Design, Dynamic Systems, Mechatronics, Feedback Systems

Awards & Honors: Dean's List, Tau Beta Pi, Pi Tau Sigma, RISE Germany scholarship

RELEVANT EXPERIENCE

Hyphen **Jun. 2026 - Sept. 2026**

Systems and Test Engineering Intern (Incoming)

San Jose, CA

Northwestern University Space Technology and Rocketry Society

Sept. 2023 - Present

Chief Engineer (Incoming 2026-2027)

Evanston, IL

Launch Vehicle Lead

- Led design for the International Rocketry Engineering Competition 2026 team rocket, 10k ft altitude category
- Developed modular fin system for Student Launch 2025, presenting design reviews to NASA technical experts
- Fabricated carbon fiber wound coupling element to accommodate 20% larger payload airframe diameter
- Earned NAR Level 2 High Power Rocketry certification in 2025 with J-class impulse motor

German Aerospace Center

Jun. 2025 - Aug. 2025

Engineering Research Intern

Stuttgart, Germany

- Conducted thermal analysis of 18+ self-prepared thermoplastic samples with Differential Scanning Calorimetry
- Analyzed user-influenced variability to improve integral bound-selection and reduce enthalpy uncertainty by 15%
- Automated analysis of 300+ thermal heating curves using Python (NumPy, Pandas, SciPy, Matplotlib libraries)

Meemo

Jul. 2025 - Sept. 2025

Industrial Design Intern

Remote

- Engineered two adjustable strap configurations to support ergonomic fit of distance trackers for parent-child users
- Modified prototype device housing shell using SolidWorks to integrate haptics motor, speaker and PCB
- Applied Design for Assembly principles to produce CAD and assembly plans for prototype manufacturing

Center for Robotics and Biosystems, Swarm Robotics Lab

Mar. 2024 - May 2026

Undergraduate Research Assistant

Evanston, IL

- Developed OptiTrack visual-based program in Python/C supporting multi-landing sequence for 50 aerial robots
- Conducted flight experiments, tuning PID/LQR controllers to improve stability and position accuracy to 2 cm
- Designed and prototyped lightweight PLA mechanical housings of under 2.3 grams to mount onboard electronics

Activities: NU Society of Women Engineers Co-President (2025-2026), Dearborn Observatory Tour Guide

PROJECTS

Quadrotor Design & Autonomous Control **Spring quarter, 2025**

- Built quadrotor robot, integrating sensing, control, and actuation in a 10-week collaborative project
- Developed autonomous PID control using IMU and camera feedback, with joystick override for robust operation
- Achieved stable hover and ground effect tag-following under full autonomous control

Bike Brake Calipers

Spring quarter, 2025

- Defined key performance metrics using ISO standards to guide project work in a team of five
- Produced CAD models with topology optimization to prepare 2 prototype iterations on Formlabs Fuse 1+ printer
- Reduced caliper weight by 5% from initial design and successfully stopped the bike within 12.4 m safety distance

SKILLS

Programming: Python, C, MATLAB, Linux (CLI), Git, I2C, SPI, UART

CAD: OnShape, Siemens NX, SolidWorks

Mechanical Prototyping: 3D printing (FDM), UTM testing, Mill, GD&T, Workshop tools

Language: English (Native), French (Fluent), German (Beginner)